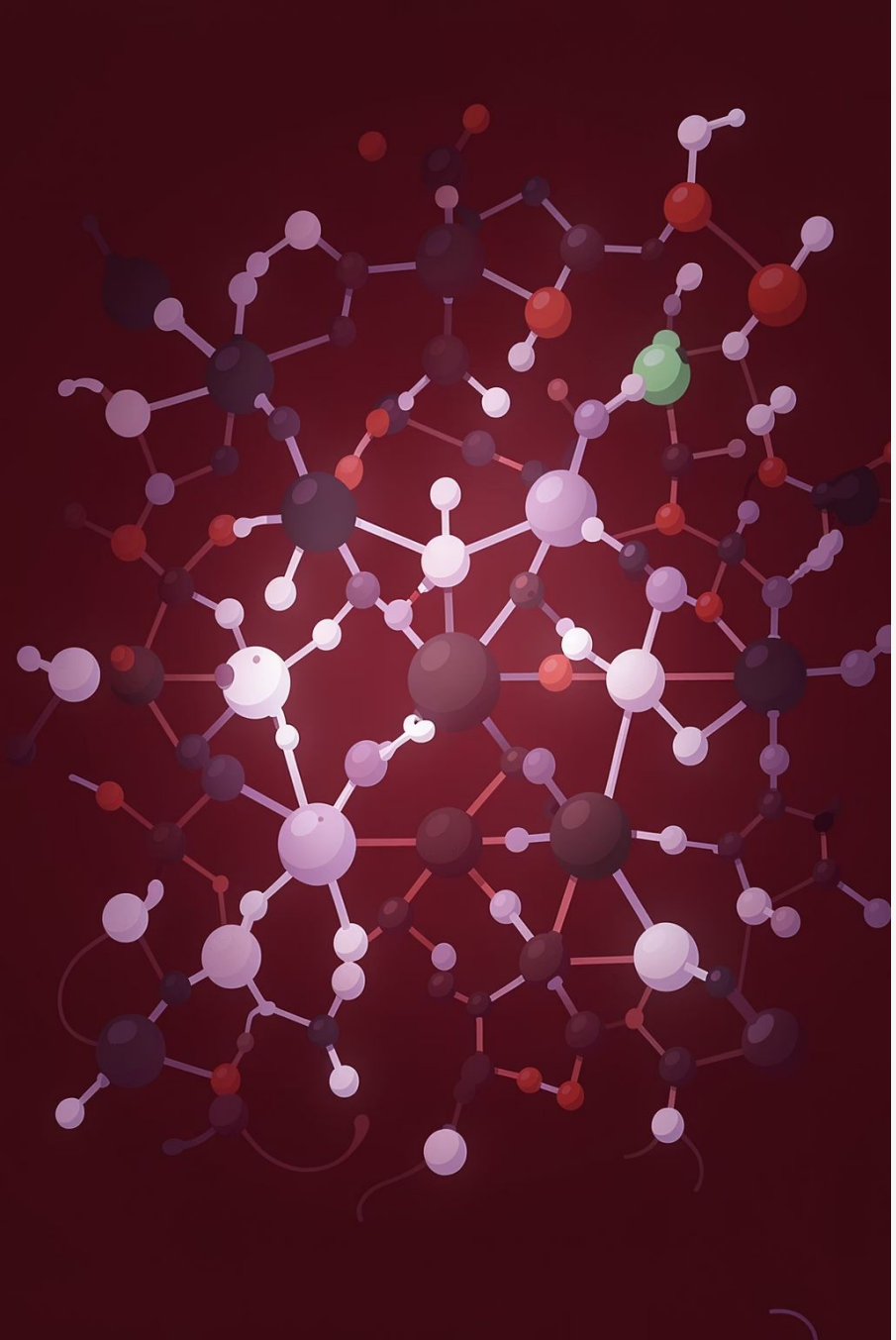


Frank's Model

The Origin of Biological Homochirality: Frank's Autocatalytic Model

How can a symmetric world produce asymmetric life?



by Saswata Chatterjee

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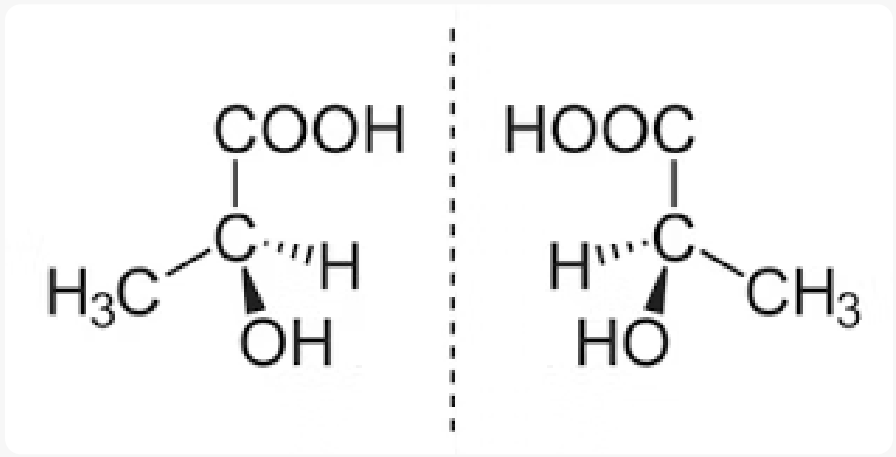
Chirality, Enantiomers and Racemic Mixtures

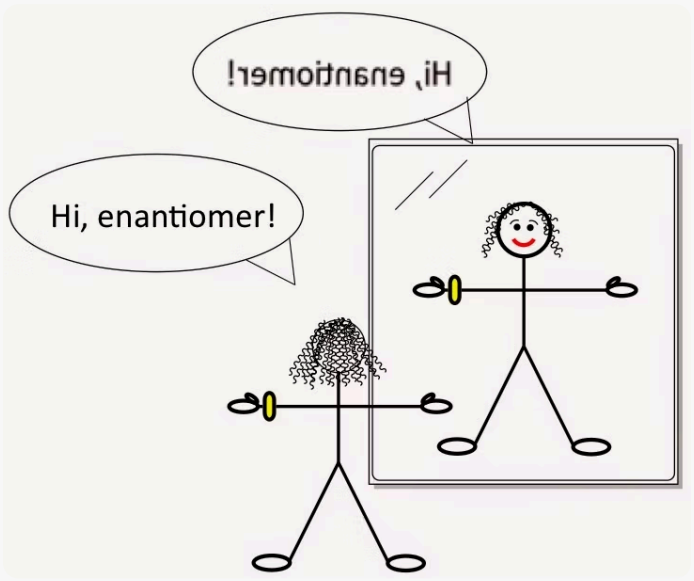
Chirality

- A chiral molecule **cannot** be superimposed on its mirror image.
- Its mirror-image forms are called **enantiomers**.
- They are often labeled **L** (left-handed) and **D** (right-handed).
- **Amino Acids:** Nearly exclusively **L-chiral**.
- **Sugars:** Nearly exclusively **D-chiral**.

Racemic Mixture

- Without a chiral influence, reactions usually form both enantiomers equally.
- A **50:50 mix** of L and D enantiomers is a **racemic mixture**.
- It has **no overall chirality**.





These two molecules are **Enantiomers** as they are **non super-imposible** and **mirror images**

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Frank's Original Model (1953)

Frank proposed the first theoretical model explaining how **spontaneous symmetry breaking** can produce homochirality through autocatalysis and mutual antagonism.

Reaction Scheme

Autocatalytic Production for D

$$A + D \xrightarrow{k_a} 2D$$

Autocatalytic Production for L

$$A + L \xrightarrow{k_a} 2L$$

Mutual Antagonism

$$D + L \xrightarrow{k_i} \varnothing$$

Key Assumptions

- The achiral precursor concentration is held constant: $[A] = a = \text{constant}$
- The system is treated as an **open system**, with fresh A supplied externally.
- Opposite enantiomers inhibit each other through **mutual antagonism**, leading to amplification of an initial chiral imbalance.

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Limitation of Frank's Original Model

Although Frank's model successfully demonstrates **spontaneous symmetry breaking**, it does **not** possess a finite steady state.

Problem Statement

The achiral precursor is assumed to remain constant:

$$[A] = a = \text{constant}$$

This means fresh A is continuously supplied, allowing autocatalysis to proceed indefinitely.

Asymptotic Behavior

As the system approaches homochirality:

$$L \to 0$$

the dominant enantiomer satisfies:

$$\frac{dD}{dt} \approx k_a a D.$$

Rate Equations

The rate equations are:

$$\frac{dD}{dt} = k_a a D - k_i D L$$

$$\frac{dL}{dt} = k_a a L - k_i D L$$

In this limit, the mutual antagonism term becomes negligible, and the dominant species grows according to simple exponential dynamics.

Key Result

$$D(t) = D_0 e^{k_a a t}$$

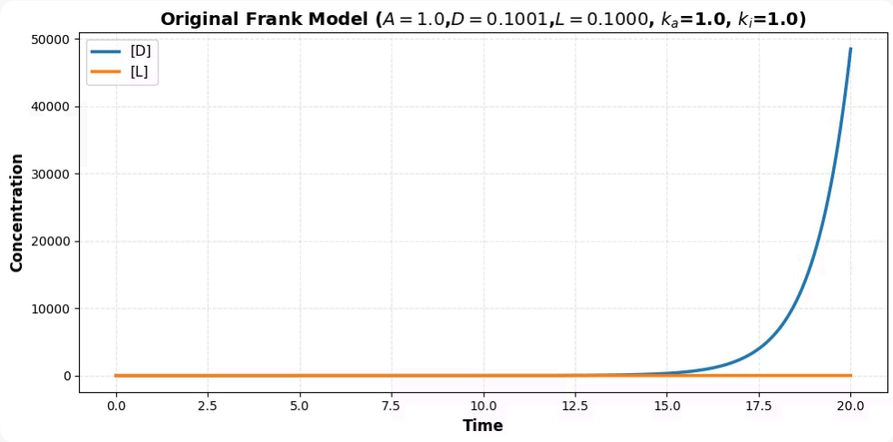
This growth is unbounded. The model therefore cannot reach a finite equilibrium concentration.

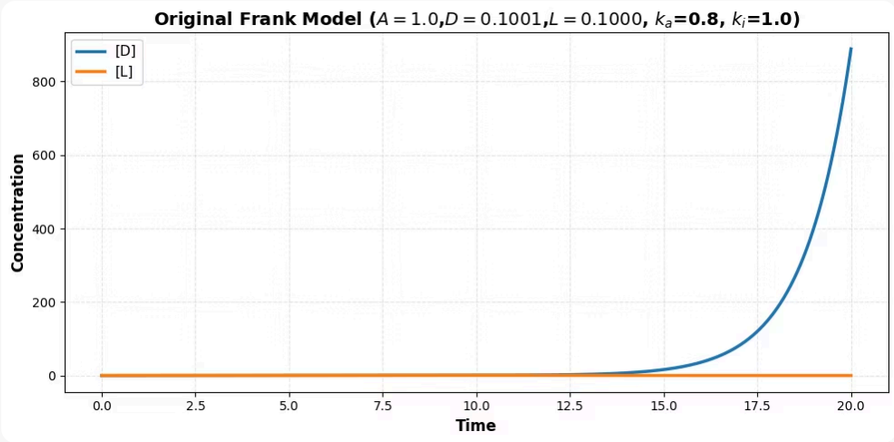
Consequence

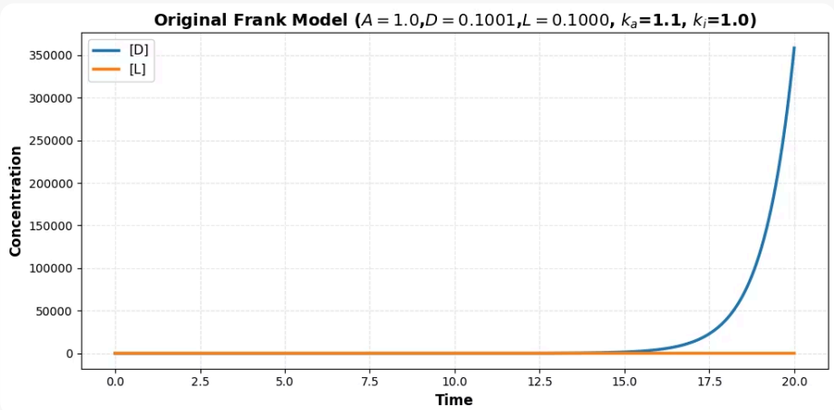
- Continuous external supply of A sustains autocatalysis indefinitely.
- The dominant enantiomer never reaches a finite equilibrium concentration.
- The model therefore has **no well-defined steady state**, motivating later molecule-conserving modifications.

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Some key results:







Conclusion:

- The original Frank model is highly unrealistic as there is no fixed steady state for the system and there is an open system an a potentially infinite supply for A molecules.
- We look at a modified reaction scheme for the Frank Model.

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Why Not Simply Let A Vary?

Suppose Frank's original reactions are retained but the assumption $[A] = \text{constant}$ is removed.

- The autocatalytic reactions consume the achiral precursor:

$$A + D \rightarrow 2D, \qquad A + L \rightarrow 2L.$$

- However, the inhibition reaction $D + L \rightarrow \varnothing$ permanently removes molecules from the system. There will be a point of time the concentration of A and L will both be 0 as A is never replinished.
- The total number of molecules is therefore **not conserved**:

$$\frac{d}{dt}(A + D + L) = -2k_i DL < 0$$

- If one attempts to formulate Frank's original reaction scheme as a closed system by allowing A to vary, the reaction violates conservation of mass because molecules disappear from the system..

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Frank's Model of Homochirality (Modified)

Frank proposed that chiral molecules are produced autocatalytically from an achiral precursor A.

Autocatalytic Production

$$A + D \xrightarrow{k_a} 2D$$

$$A + L \xrightarrow{k_a} 2L$$

where:

- A = achiral precursor
- D = D-enantiomer
- L = L-enantiomer
- k_a = autocatalytic rate constant

Chiral Inhibition

$$D + L \xrightarrow{k_i} 2A$$

where:

- k_i = chiral inhibition rate constant

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Modifications introduced in Frank Model (Jafarpour et al.)

Jafarpour *et al.* modified Frank's original reaction scheme to obtain a **well-defined steady state** while preserving the symmetry-breaking mechanism.

Original Model

- The achiral precursor concentration is held constant.
- Chiral inhibition is represented by

$$D + L \rightarrow \varnothing$$

- This creates an **open system** with no conservation of the total number of molecules.

Modified Model

- The inhibition reaction is replaced by

$$D + L \rightarrow 2A$$

- Opposite enantiomers are recycled back into the achiral precursor.
- The total number of molecules is conserved:

$$[A] + [D] + [L] = N$$

- This produces a finite, well-defined steady state and simplifies the mathematical analysis.

Key difference: the modified model closes the system by recycling D and L into A, enabling a stable steady state.

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Chiral Order Parameter

The state of the system is described by the chiral order parameter

$$\omega = \frac{[D] - [L]}{[D] + [L]}$$

Interpretation

- $\omega = 0 \rightarrow$ Racemic mixture
- $\omega = +1 \rightarrow$ Pure D state
- $\omega = -1 \rightarrow$ Pure L state

Objective

Explain how a system starting near

$$\omega \approx 0$$

can evolve toward

$$|\omega| = 1$$

through spontaneous symmetry breaking.

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Mean-Field Dynamics of Frank's Model

The reactions in Frank's model can be translated into a set of deterministic **mean-field equations** using mass-action kinetics.

Mean-Field Equations

$$\frac{d[A]}{dt} = 2k_i[D][L] - k_a[A]\big([D]+[L]\big)$$

$$\frac{d[D]}{dt} = k_a[A][D] - k_i[D][L]$$

$$\frac{d[L]}{dt} = k_a[A][L] - k_i[D][L]$$

Parameters

- $[A]$: concentration of the achiral precursor
- $[D]$: concentration of the D-enantiomer
- $[L]$: concentration of the L-enantiomer
- k_a : autocatalytic rate constant
- k_i : chiral inhibition rate constant

Evolution of the Order Parameter

Using the mean-field equations, the evolution of ω is

$$\frac{d\omega}{dt} = \frac{1}{2}k_i([D]+[L])\,\omega\,(1-\omega^2)$$

Key Result

The factor

$$\omega(1-\omega^2)$$

produces three fixed points:

$$\omega = -1, \quad 0, \quad +1$$

representing two homochiral states and one racemic state. This three-fixed-point structure is the mathematical basis of Frank's spontaneous symmetry-breaking mechanism.

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Stability Analysis

Understanding whether an equilibrium is stable, unstable, or a saddle point helps determine how a dynamical system responds to small perturbations.

Fixed Point

A fixed point is a state where the system no longer evolves.

$$\frac{dD}{dt} = 0, \qquad \frac{dL}{dt} = 0$$

At a fixed point, the concentrations remain constant unless the system is perturbed.

⚠ Nearby trajectories reveal whether the equilibrium attracts, repels, or behaves differently along distinct directions.

Equilibrium Criteria

Stable Equilibrium

Nearby trajectories return to the fixed point after a small perturbation.

Interpretation: the equilibrium attracts nearby states.

Unstable Equilibrium

Nearby trajectories move away from the fixed point after a small perturbation.

Interpretation: the equilibrium repels nearby states.

Key Takeaway

By combining the fixed-point condition with local stability analysis, we can determine whether an equilibrium is stable, unstable, or a saddle point.

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Linear Stability Analysis :

To characterize the stability of the racemic equilibrium, the conservation law

$$A + D + L = N$$

was first used to eliminate the achiral precursor,

$$A = N - D - L$$

reducing the system to two coupled nonlinear ODEs in D and L. The racemic equilibrium was then obtained by imposing:

$$\dot{D} = \dot{L} = 0$$

By analysing the nullclines we can understand the equilibrium points which would be located at their intersection. However it would reveal results which are not important or trivial , thus together with the symmetry condition $D = L$. We can obtain points which form a **Racemic Equilibrium**.

$$\boxed{D^* = L^* = \frac{k_a N}{2k_a + k_i}, \qquad A^* = \frac{k_i N}{2k_a + k_i}.}$$

For the symmetric case $(k_a = k_i)$:

$$\boxed{A^* = D^* = L^* = \frac{N}{3}.}$$

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Jacobian and Eigenvalue analysis:

The reduced system was linearized about the racemic equilibrium by constructing the Jacobian matrix for these equations:

$$f(D,L) = k_a D(N - D - L) - k_i DL$$

$$g(D,L) = k_a L(N - D - L) - k_i DL$$

The Jacobian is:

$$J = \begin{pmatrix} \dfrac{\partial f}{\partial D} & \dfrac{\partial f}{\partial L} \\ \dfrac{\partial g}{\partial D} & \dfrac{\partial g}{\partial L} \end{pmatrix}$$

Evaluating at the racemic equilibrium gives:

$$J = -\frac{k_a N}{2k_a + k_i} \begin{pmatrix} k_a & k_a + k_i \\ k_a + k_i & k_a \end{pmatrix}$$

The corresponding eigenvalues are:

$$\boxed{\lambda_1 = -k_a N, \qquad \lambda_2 = \frac{k_a k_i N}{2k_a + k_i}}$$

with the eigen vectors:

$$\boxed{\mathbf{v}_1 = (1,1)^T, \qquad \mathbf{v}_2 = (1,-1)^T}$$

Interpretation

- Since k_a, k_i, N are all positive quantities inherently it is fair to say that one of my eigven values is negative while the other is positive.
- Under these conditions all implications lead to my equilibrium point being a Saddle Point as $\lambda 1 < 0, \lambda 2 > 0$. This implies our equilibrium is linearly unstable with perturbations growing along one directions and decaying along the other.
- The racemic equilibrium lies on the symmetry line $\boxed{D = L}$

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Assumptions for Phase Space Analysis

To simplify the qualitative analysis of the modified Frank model, the following assumptions are made:

> For analytical tractability, we impose the symmetry condition $D = L$to isolate the racemic branch of equilibria, rather than solving the full nullcline system for arbitrary $k_a \neq k_i$

The total concentration is normalized

$$A + D + L = 1$$

which is equivalent to scaling the conserved quantity N by the total number of molecules- equal reaction rates are assumed,

$$k_a = k_i = 1$$

to obtain a **symmetric system** where autocatalysis and chiral inhibition are equally weighted. This normalization does not alter the qualitative dynamics or stability of the system, but simplifies the analytical treatment.

Under these assumptions,

The nullcline equations obtained are:

$$\dot{D} = D(1 - D - 2L) = 0$$

$$\dot{L} = L(1 - 2D - L) = 0$$

Solving these, we get 4 points of intersection:

$$\boxed{\left\{(0,0),\ (1,0),\ (0,1),\ \left(\frac{1}{3}, \frac{1}{3}\right)\right\}}$$

The second point which can also be obtained by just computing the parameters in the solved D and L value before is:

$$\boxed{(D^*, L^*) = \left(\frac{1}{3}, \frac{1}{3}\right).}$$

As discussed earlier this is our **Saddle Point.**

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Jacobian Analysis at all these points:

For the normalized modified Frank model,

$$A + D + L = 1, \qquad k_a = k_i = 1$$

The Jacobian matrix is:

$$J(D, L) = \begin{pmatrix} 1 - 2D - 2L & -2D \\ -2L & 1 - 2D - 2L \end{pmatrix}$$

Now we can individually discuss the stability of all of the 4 points which were previously obtained.

---

1. Achiral Equilibrium

$$(D, L) = (0, 0)$$

Jacobian

$$J(0, 0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Eigenvalues

$$\lambda_1 = \lambda_2 = 1$$

Eigenvectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 & 0 \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} 0 & 1 \end{pmatrix}.$$

Stability

Both eigenvalues are positive.

Unstable Source

2. Racemic Equilibrium

$$(D, L) = \left(\frac{1}{3}, \frac{1}{3}\right)$$

Jacobian

$$J = \begin{pmatrix} -\frac{1}{3} & -\frac{2}{3} \\ -\frac{2}{3} & -\frac{1}{3} \end{pmatrix}$$

Eigenvalues

$$\lambda_1 = -1, \qquad \lambda_2 = \frac{1}{3}$$

Eigenvectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 & 1 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 & -1 \end{pmatrix}$$

Stability:

Saddle Point

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3. D-Homochiral Equilibrium

$$(D, L) = (1, 0)$$

Jacobian

$$J = \begin{pmatrix} -1 & -2 \\ 0 & -1 \end{pmatrix}$$

Eigenvalues

$$\lambda_1 = \lambda_2 = -1$$

Eigenvector

$$\mathbf{v} = \begin{pmatrix} 1 & 0 \end{pmatrix}$$

The Jacobian is **defective**, possessing only one linearly independent eigenvector.

Stability: Stable Improper Node

4. L-Homochiral Equilibrium

$$(D, L) = (0, 1)$$

Jacobian

$$J = \begin{pmatrix} -1 & 0 \\ -2 & -1 \end{pmatrix}$$

Eigenvalues

$$\lambda_1 = \lambda_2 = -1$$

Eigenvector

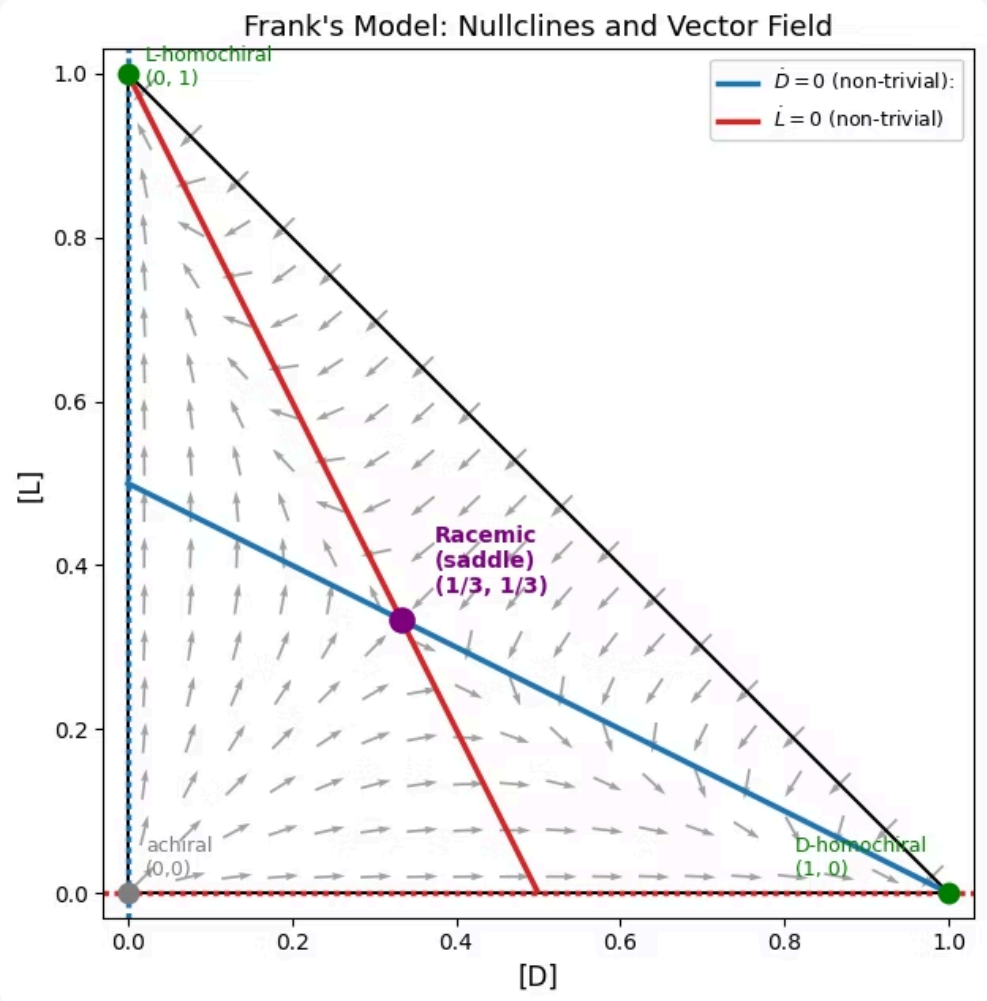
$$\mathbf{v} = \begin{pmatrix} 0 & 1 \end{pmatrix}$$

The Jacobian is **defective**, possessing only one linearly independent eigenvector.

Stability: Stable Improper Node

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Analysis of the Phase Space:



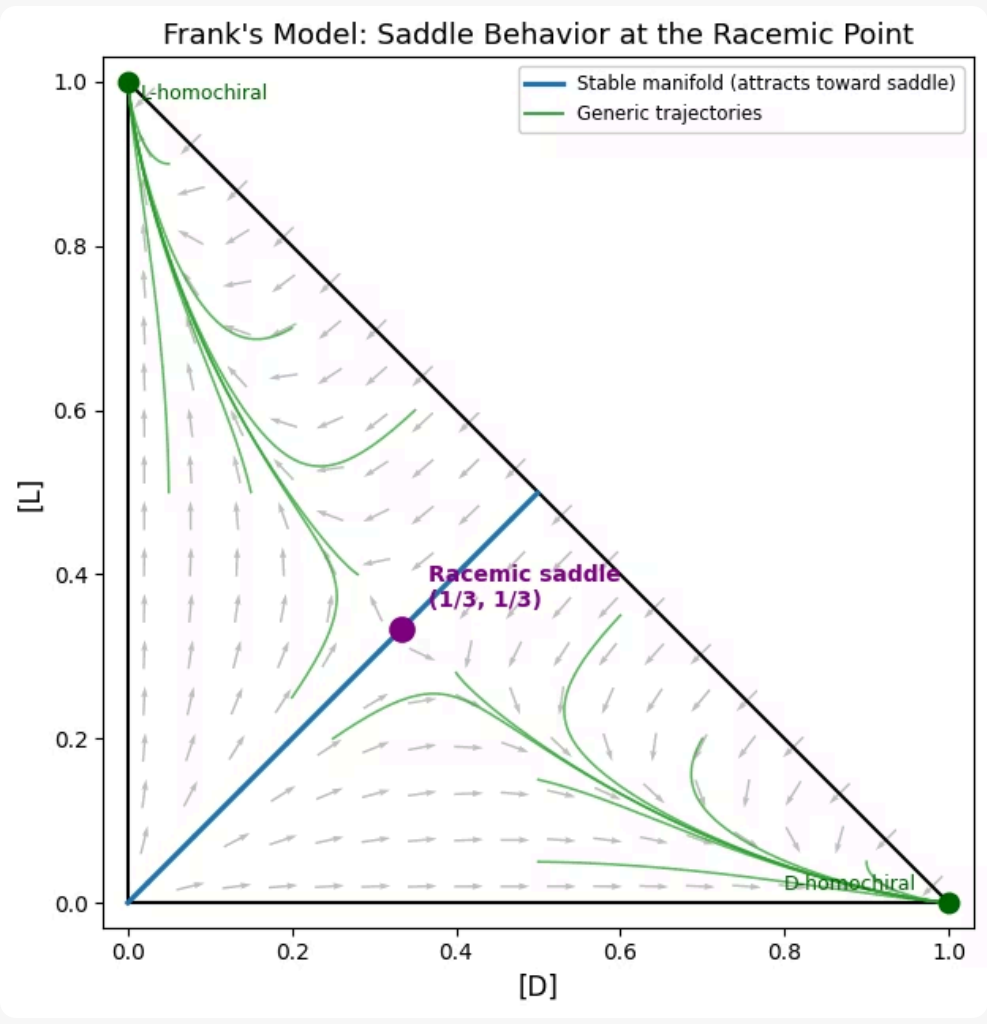
The figure along side is a vector field with the D and L nullcline, there are some interesting inferences we can draw from it:

- The Jacobian eigenanalysis predicted the racemic equilibrium to be a saddle point:

$$\left(\frac{1}{3}, \frac{1}{3}\right)$$

this perfrectly follows our calculation.

- The vector field gives us a great picture which allows us to visualise the Saddle point at our racemic equilibrium.



This figure is specifically illustrating the **saddle nature** of the racemic equilibrium:

- This behaviour is consistent with the analytical eigenvalue analysis
- The stable manifold coincides with the symmetry line

$$D = L$$

corresponding to the eigenvector

$$v_1 = (1, 1)^T$$

- Perturbations perpendicular to this manifold correspond to the unstable eigenvector

$$v_2 = (1, -1)^T$$

which amplifies an initially small enantiomeric imbalance and drives spontaneous symmetry breaking.

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Relating Phase Space to the Order Parameter

Although the stability analysis was performed in the (D, L) phase space, it is entirely consistent with the order parameter ω.

We had previously stated our enantiomeric excess order parameter as:

$$\omega = \frac{D - L}{D + L}$$

The symmetry line $D = L$ corresponds to $\boxed{\omega = 0}$

Two key observations which could be made from our previous calculations are:

1. The saddle behaviour observed at

$$\left(\frac{1}{3}, \frac{1}{3}\right)$$

shows that infinitesimal perturbations away from $D = L$ grow with time, causing

$$\omega = 0$$

to become unstable.

2. As trajectories evolve toward either homochiral equilibrium,

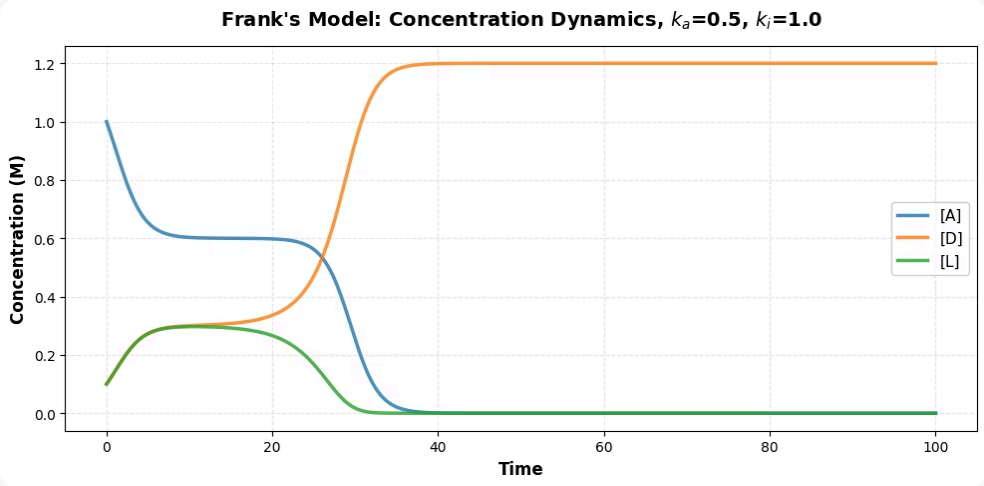
$$\omega \to \pm 1$$

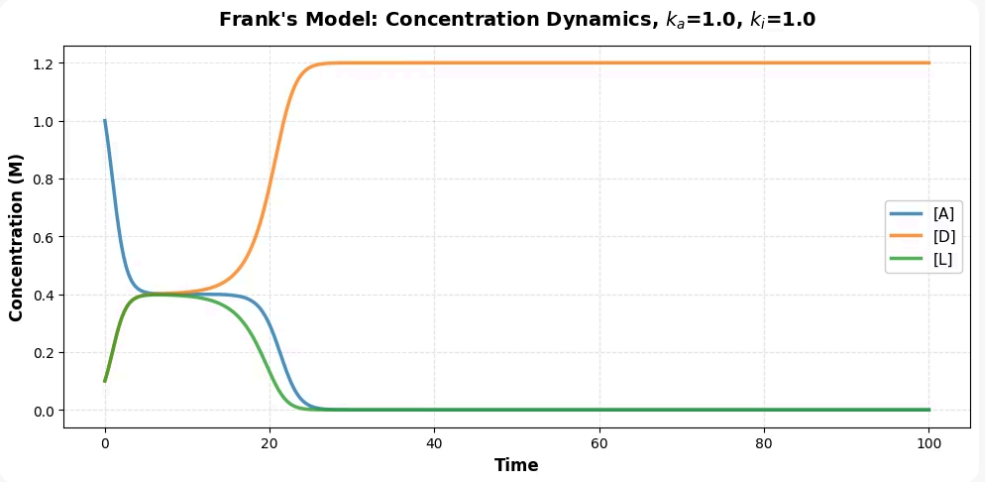
confirming the stable fixed points predicted by the analytical solution of the order parameter equation.

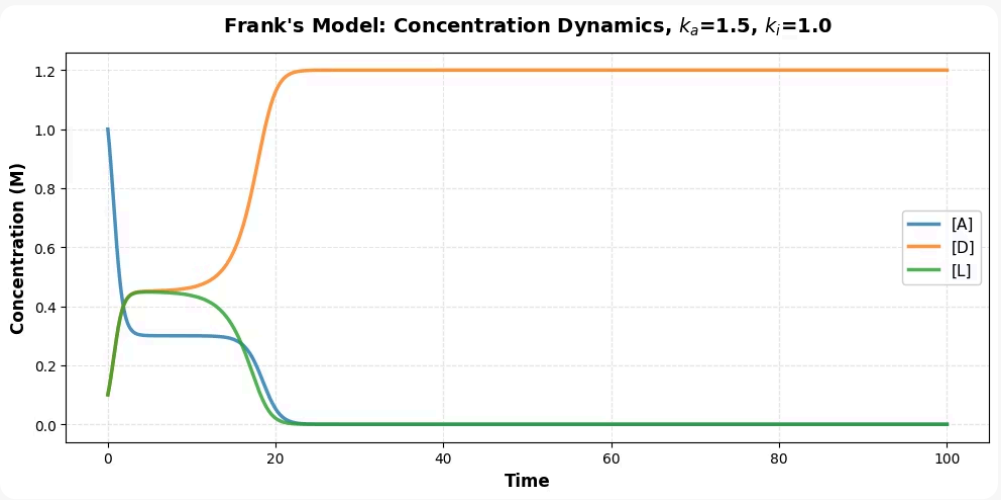
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Deterministic Results

1. $[A_0]$=1.0 , $[D_0]$=0.1001 , $[L_0]$=0.1000, these are the starting concentrations of each speices:







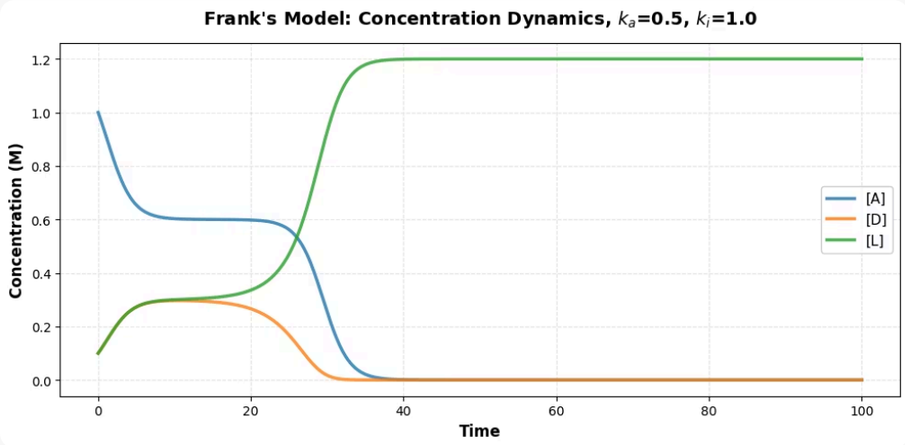
Conclusion

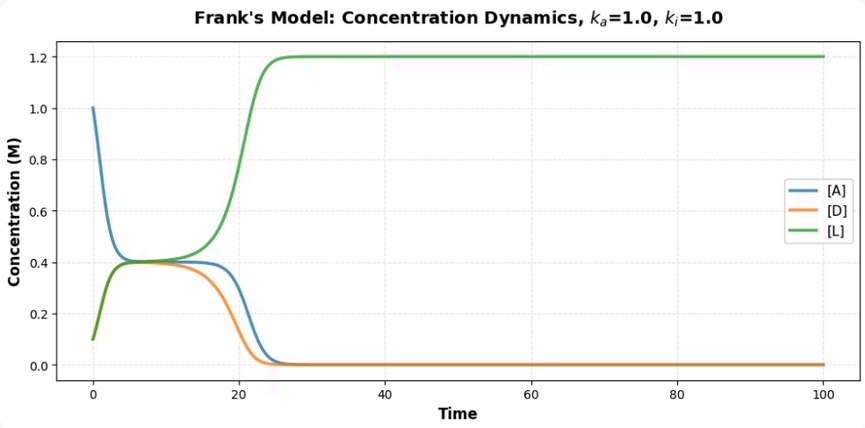
- Even a very small initial dominance is enough for the system to amplify it over time to a state of homochirality
- The k_a: k_i ratio is used to determine how early the leading species take dominance however it cannot change the final outcome

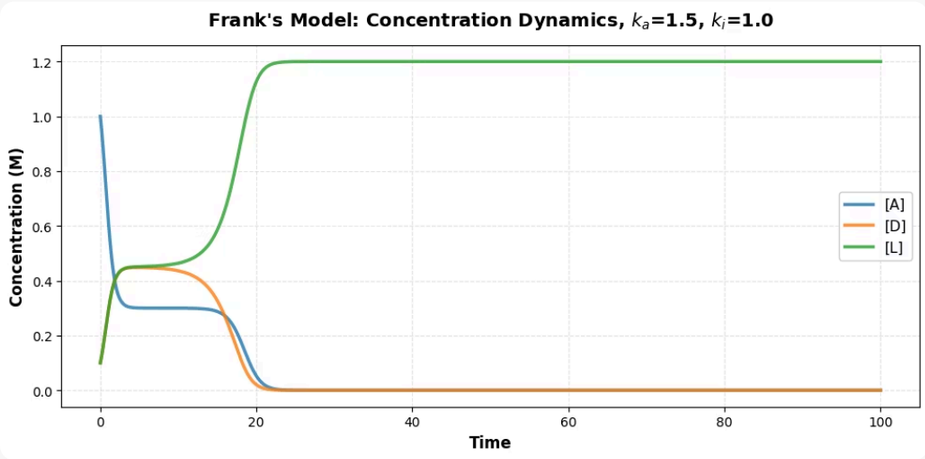
⚠ **NOTE:** The the system drives toward homochirality but **mathematically never reaches it in finite time**. The minority enantiomer decays exponentially toward zero but asymptotically — it takes infinite time to reach exactly zero. So deterministically you get *arbitrarily close* to homochiral but never 100%

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2.[A_0]=1.0 , [D_0]=0.1000, [L_0]=0.1001, these are the starting concentrations of each speices:





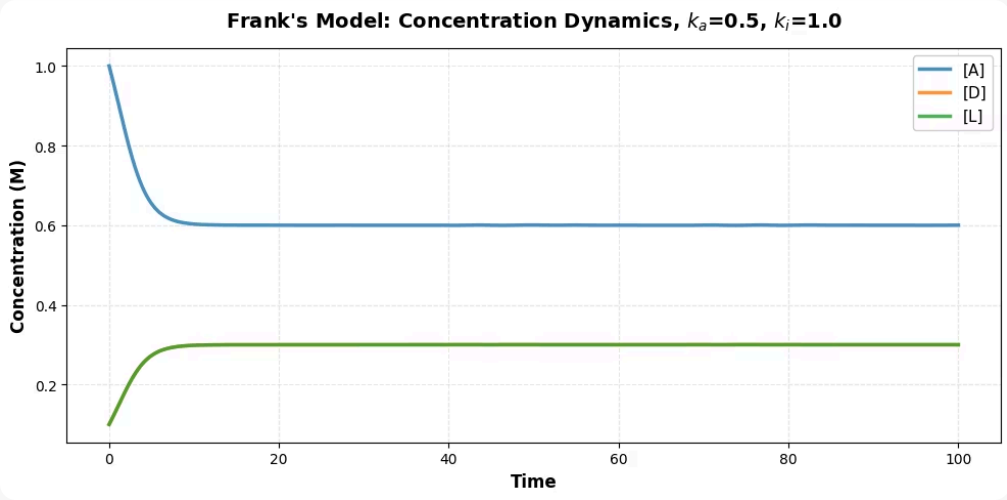


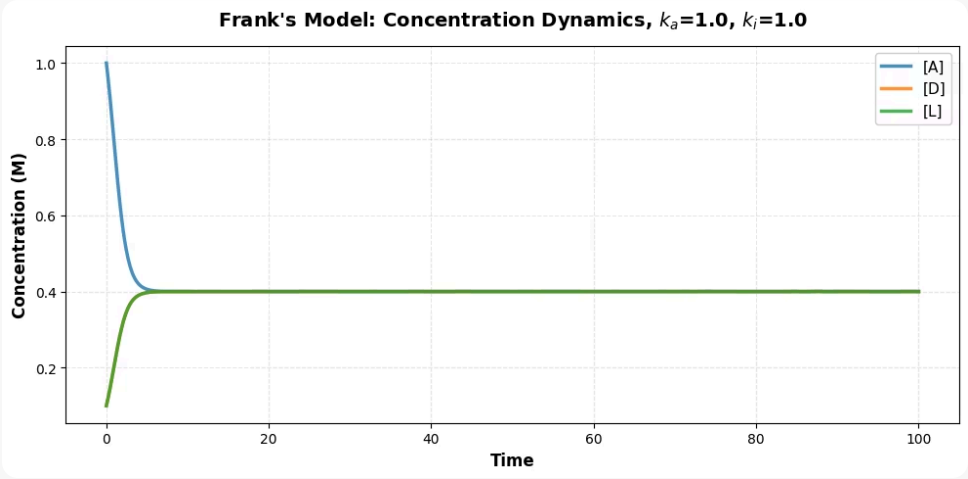
Conclusion:

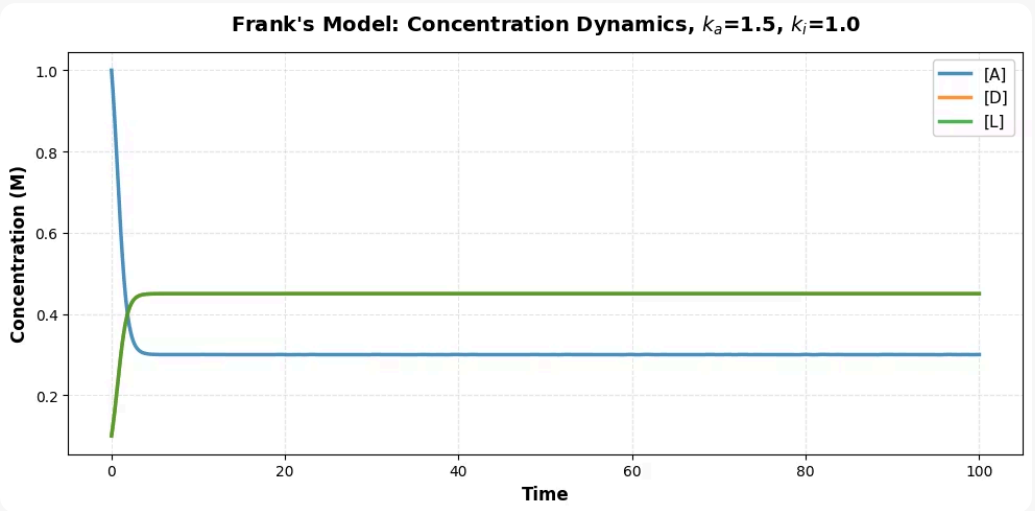
- As expected the L specie at the beginning had a slight edge and it gets amplified to a point of homochirality at the end.
- The k_a: k_i ratio is again only used to determine how fast the species take dominance and not which is the homochiral spieces.

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3.$[A_0]$=1.0 , $[D_0]$=0.1000 , $[L_0]$=0.1000, these are the starting concentrations of each speices:







Conclusion:

- The initial species being a racemic mixture yield a racemic mixture as there is no perturbation in the system to amplify it.
- If we consider an initial constraint that

$$A + D + l = N$$

At equilibrium, the production and inhibition rates of both enantiomers must balance. Since $k_a = k_i$, the equilibrium conditions reduce to

$$A = D = L = N/3$$

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Gillespie Simulation

Since the deterministic model of Frank's Model failed to break the homochirality of the species we wish to examine the stochastic behaviour of our system with the help of Gillespie Simulation. We individually run singular trajectories and analyse something really interesting :

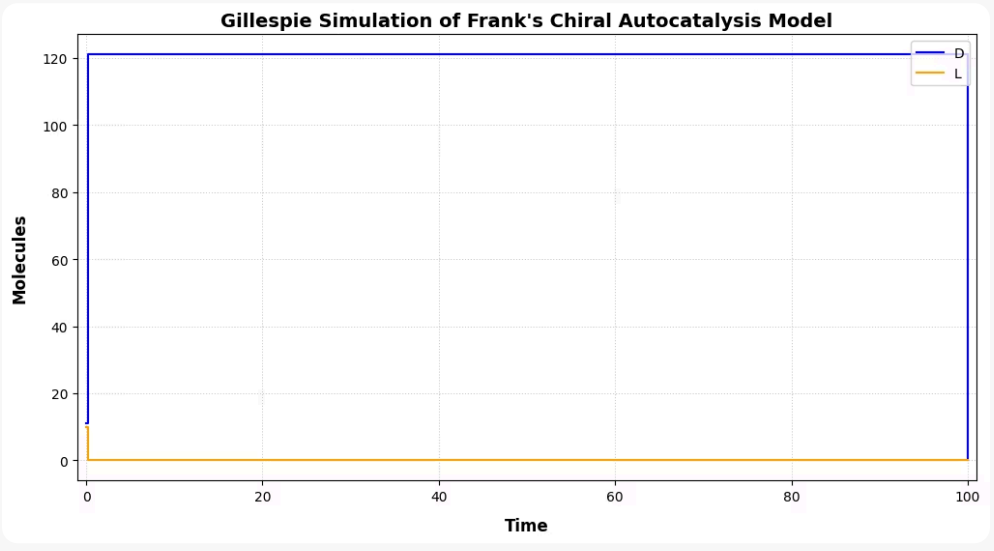
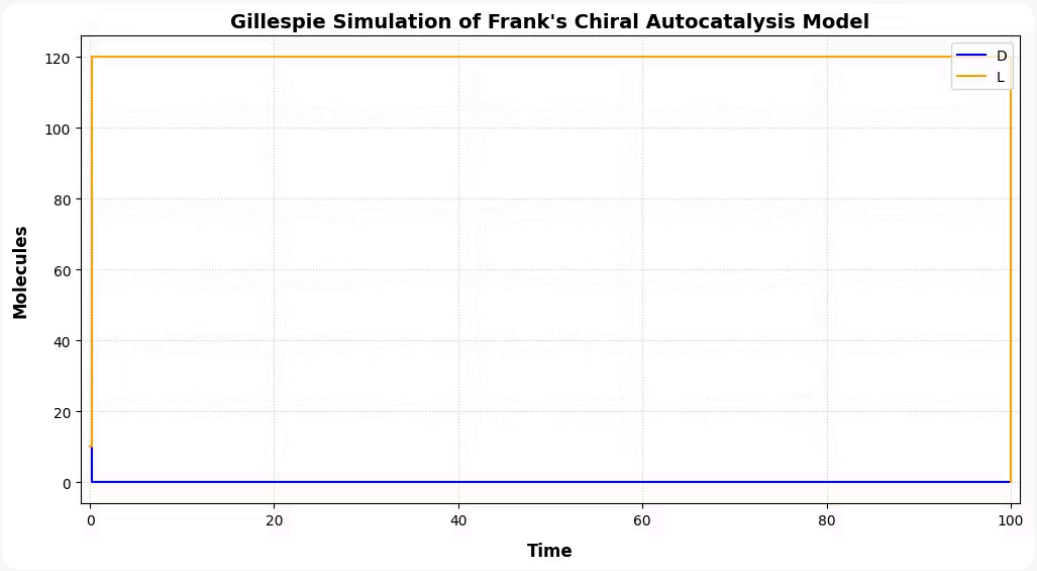


Diagram details:

- k_a = k_i =1.0 (highly inconsequential in determining the dominant species here which matches with the deterministic model).
- A_0 =100, D_0 =10, L_0 =10
- Both trajectories have the same parameter with absolutely no starting bias



Observation:

This is again an identical simulation which results in a completely different outcome where L is the dominant speices. Running numerous simulations yeilded similar results where the winner was unpredictable.

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Statistical Validation of Stochastic Symmetry Breaking

Null hypothesis:

$$H_0: \quad P(D) = P(L) = 0.5$$

Interpretation: The Gillespie implementation is statistically consistent with the theoretical symmetry of the modified Frank model.

Alternative hypothesis:

$$H_1: \quad p \neq 0.5$$

Interpretation: The Gillespie implementation exhibits a measurable bias toward one homochiral state.

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Methodology used for validation:

1. Perform (n=500) independent Gillespie simulations.
2. Define the outcome of the (i^{th}) trajectory as

$$X_i = \begin{cases} 1, & \text{if trajectory } i \text{ reaches the } D\text{-homochiral state,} \\ 0, & \text{if trajectory } i \text{ reaches the } L\text{-homochiral state.} \end{cases}$$

3. Each trajectory is therefore an independent Bernoulli random variable

$$X_i \sim \text{Bernoulli}(p)$$

4. The total number of trajectories ending in the (D)-homochiral state is

$$K = \sum_{i=1}^{n} X_i$$

which follows

$$K \sim \text{Binomial}(n, p)$$

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5. **Maximum Likelihood Estimate**

The probability of reaching the (D)-homochiral state is estimated using the Maximum Likelihood Estimator

$$\hat{p} = \frac{K}{n}$$

6. For sufficiently large (n), the Central Limit Theorem gives

$$\hat{p} \sim N\left(p, \frac{p(1-p)}{n}\right)$$

Since the true probability (p) is unknown, it is approximated using the MLE,

$$SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

7. Statistical Validation: Construct the 95% confidence interval

$$\hat{p} \pm 1.96SE$$

Perform an exact two-sided binomial test under

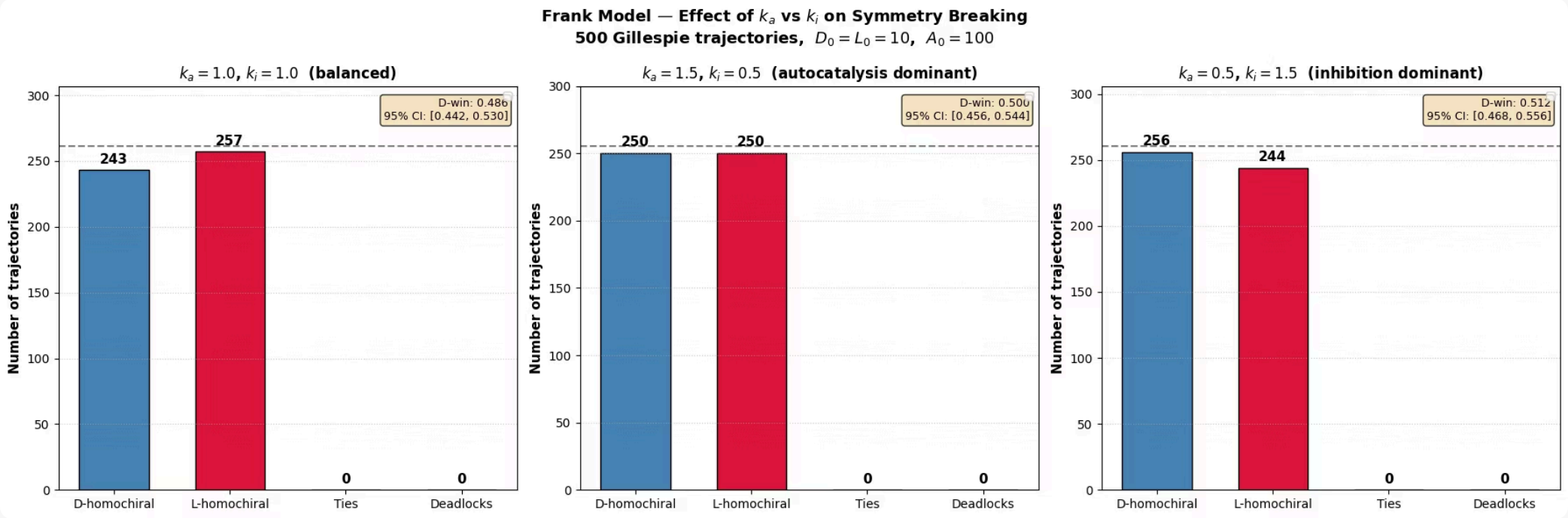
8.

$$H_0 : \ p = 0.5$$

If the confidence interval contains (0.5) and the binomial test does not reject $H_0 : \ p = 0.5$ the observed Gillespie simulations are considered **statistically consistent** with the theoretical prediction of unbiased stochastic symmetry breaking.

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Results:



1. D-homochiral outcomes: **243**
2. L-homochiral outcomes: **257**
3. MLE probability: $\hat{p} = 0.486$
4. 95% Confidence Interval:

$[0.442,\ 0.530]$

1. D-homochiral outcomes: **250**
2. L-homochiral outcomes: **250**
3. MLE probability: $\hat{p} = 0.5004$. 95%
4. 95% Confidence Interval:

$[0.456,\ 0.544]$

1. D-homochiral outcomes: **256**
2. L-homochiral outcomes: **244**
3. MLE probability: $\hat{p} = 0.512$
4. 95% Confidence interval:

$[0.468,\ 0.556]$

Interpretation: Across all three parameter regimes, the observed proportions of D- and L-homochiral outcomes remain statistically consistent with the theoretical prediction $P(D) = P(L) = 0.5$, as the value 0.50 lies within every 95% confidence interval. Consequently, there is insufficient statistical evidence to reject the null hypothesis, indicating that variations in k_a and k_i influence the reaction dynamics but do not introduce a measurable chiral bias. These results support the conclusion that symmetry breaking in the modified Frank model remains an unbiased stochastic process.

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Some points to address regarding the Modified Frank Model

- The modified Frank model provides a mathematically consistent and molecule-conserving framework for studying **spontaneous symmetry breaking** and **homochirality**.
- However, it remains an **idealized theoretical model** and does not describe the detailed molecular mechanism of any specific chemical reaction.
- For over four decades, no experimental chemical system exhibiting **asymmetric autocatalysis** and **enantiomeric excess (EE) amplification** was known.
- The discovery of the **Soai reaction (1995)** provided the first experimental realization of these phenomena, offering strong experimental support for Frank's theoretical framework.
- Consequently, the modified Frank model is widely used as a **minimal theoretical model**, while the Soai reaction serves as its most significant experimental counterpart. However the issue is this was the only experimental evidence regarding the existance and real world significance of this model.
- Recent ribozyme-mediated RNA replication experiments suggest that **homochirality can emerge even in the absence of explicit chiral inhibition**, motivating the study of alternative Frank-type models beyond the original formulation.
- Jafarpour *et al.* replaced the chiral inhibition reaction with a **molecule-conserving recycling step**, converting opposite enantiomers back into the achiral precursor while preserving the essential symmetry-breaking dynamics.

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References

- **Frank, F. C. (1953).**

> Frank, F. C. *On spontaneous asymmetric synthesis*. Biochimica et Biophysica Acta, **11**, 459–463.

- **Jafarpour et al.**

> Jafarpour, F., Biancalani, T., & Goldenfeld, N. (2015). *Noise-induced mechanism for biological homochirality*. Physical Review Letters.

- **Soai Reaction**

> Soai, K., Shibata, T., Morioka, H., & Choji, K. (1995). *Asymmetric autocatalysis and amplification of enantiomeric excess*. Nature.

Software

> Python 3.13

Python was extensively used for all calculations and producing every image and statistic behind this presentation.

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Future Directions:

Going forward we will analyse the model that was newly proposed by **Jafarpour et al.**

Thank You!

